

BACKGROUND PAPER

Carbon Pricing through Emissions Trading in Asia and the Pacific: Comparative Analysis and Way Forward

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Carbon Pricing through Emissions Trading in Asia and the Pacific: Comparative Analysis and Way Forward

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Note: In this report, “\$” refers to United States dollars.

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ABBREVIATIONS

ACCU	-	Australian Carbon Credit Unit
ADB	-	Asian Development Bank
AUD	-	Australian dollar
CIPP	-	Comprehensive Investment and Policy Plan
CCR	-	cost containment reserve
CNY	-	Chinese yuan
CO ₂	-	carbon dioxide
CO ₂ e	-	carbon dioxide equivalent
CRF	-	Common Reporting Format
EDGAR	-	Emissions Database for Global Atmospheric Research
EITE	-	emissions-intensive-trade-exposed
ETS	-	emissions trading system
EU	-	European Union
EUA	-	EU Allowance
GDP	-	gross domestic product
GHG	-	greenhouse gas
GtCO ₂ e	-	gigaton of carbon dioxide equivalent
GX-ETS	-	Green Transformation ETS
HFC	-	hydrofluorocarbon
ICAP	-	International Carbon Action Partnership
IDR	-	Indonesian rupiah
IEA	-	International Energy Agency
JETP	-	Just Energy Transition Partnership
KAU	-	Korean Allowance Unit
KRW	-	Republic of Korea won
KZT	-	Kazakhstani tenge
LULUCF	-	land use, land use change, and forestry
MAC	-	marginal abatement cost
MEE	-	Ministry of Ecology and Environment (PRC)
MfE	-	Ministry for the Environment (New Zealand)
MRV	-	monitoring, reporting, and verification
MSR	-	Market Stability Reserve
MtCO ₂	-	metric ton of carbon dioxide
MtCO ₂ e	-	metric ton of carbon dioxide equivalent
MW	-	megawatt

MWh	-	megawatt hour
NAP	-	national allocation plan
NEK	-	Nilai Ekonomi Karbon (Indonesia's ETS)
NZD	-	New Zealand dollar
NZU	-	New Zealand Unit
PFC	-	perfluorocarbon
PRC	-	People's Republic of China
PSAM	-	price or supply adjustment measure
PTBAE-PU	-	Persetujuan Teknis Batas Atas Emisi-Pelaku Usaha (Indonesia's allowances)
ROK	-	Republic of Korea
SF ₆	-	sulfur hexafluoride
SMC	-	Safeguard Mechanism Credit
tCO ₂ e	-	ton of carbon dioxide equivalent
TPS	-	tradable performance standard
UK	-	United Kingdom
UNFCCC	-	United Nations Framework Convention on Climate Change

ABSTRACT

Carbon pricing has emerged as a key policy instrument for reducing greenhouse gas (GHG) emissions, with carbon taxes and emissions trading as the main tools. As of June 2024, 89 national and subnational jurisdictions globally have implemented carbon pricing, covering 24% of global GHG emissions. Asian countries in particular are increasingly using emissions trading systems (ETs) to meet their climate goals. This paper provides a comprehensive review of all national and subnational ETs in the Asia and Pacific region—namely, in Australia, the People’s Republic of China (PRC), Indonesia, Japan, Kazakhstan, the Republic of Korea (ROK), and New Zealand. We also examine the European Union ET (EU ET) as a reference point since it is the oldest ET for GHGs and the largest by value of trading. Each ET presents unique characteristics but they share common challenges, such as limited sectoral coverage, weak price signals, and issues related to monitoring and enforcement.

An important contribution of the paper is an assessment framework to evaluate ETs in the four countries where adequate data is available for such analysis—namely, the PRC, Kazakhstan, the ROK, and New Zealand. The framework comprises eight criteria for evaluating an ET—contribution to GHG mitigation; cost-effectiveness of mitigation; predictability; robustness; accountability and transparency; administrative cost-effectiveness; fairness; and compatibility with national circumstances and other GHG mitigation policies. These eight criteria are further elaborated into 28 specific indicators against which the paper evaluates the ETs. Based on this analysis, we recommend policy measures such as clarifying the long-term objectives of ETs; tightening emissions caps; enhancing administrative capacity for better monitoring, reporting, and verification; and complementary policies to mitigate the distributional impacts of carbon pricing. While ETs in Asia and the Pacific remain a work in progress, they offer important lessons for countries in the region that are considering or starting to implement emissions trading.

Keywords: Greenhouse gas (GHG) mitigation, Carbon pricing, Emissions trading, Emissions trading system (ETS), Asia

I. INTRODUCTION

Carbon pricing has emerged as an important policy tool in reducing greenhouse gas (GHG) emissions. The two main carbon pricing instruments are carbon taxes and carbon trading. If well designed, carbon trading, which typically takes the form of an emissions trading system (ETS), is said to “offer triple benefits by protecting the environment, driving investment in clean technology, and raising revenue for governments to invest in low-carbon technologies” (ADB 2021, p. 127). As of June 2024, 89 national and subnational jurisdictions around the world were pricing carbon; of these, 37 were ETSs and 39 were carbon taxes (World Bank nd). These two carbon pricing instruments together cover 12.8 gigatons of carbon dioxide equivalent (GtCO₂e), which is nearly a quarter (24%) of global GHG emissions,¹ for a total value of \$104 billion.² Notably, only high- and middle-income countries have adopted carbon pricing: it covers 31% of emissions from high-income countries and 22% of emissions from middle-income countries.³

Countries in Asia, with a growing role in the global economy, are significant emitters of GHGs,⁴ and hence are increasingly using carbon pricing tools to meet their national and global commitments. The region is home to several forms of national ETS—in Australia, the People’s Republic of China (PRC), Indonesia, Kazakhstan, the Republic of Korea, and New Zealand—currently under operation. Among the many Asia countries considering a national ETS are India, Malaysia, Pakistan, Taipei, China, Thailand, and Viet Nam. The region also hosts several subnational ETS systems. The PRC has implemented subnational ETSs in two provinces (Guangdong (including Shenzhen) and Hubei) and four cities (Beijing, Chongqing, Shanghai, and Tianjin). Japan has these in two cities (Saitama and Tokyo). Both these countries’ subnational ETSs are 10–15 years old and have played an important role in preparing for the nationwide rollout.

Each of the national ETSs is different, and each presents useful lessons. Differences can be found in their ambition, scope, regulations, and efficacy. Each ETS offers crucial lessons from the perspectives of design, operation, and governance. This paper provides a comparative discussion of the ETS systems in Asia and the Pacific and analyzes how effective they have been as a policy tool in their own country so far.

¹ World Bank Carbon Pricing Dashboard. [Compliance Mechanisms: Coverage](#) (accessed 23 June 2024).

² World Bank Carbon Pricing Dashboard. [Compliance Mechanisms: Revenue](#) (accessed 23 June 2024). The bulk of this revenue (\$75 billion) is from ETSs and the rest (\$29 billion) is from carbon taxes.

³ World Bank Carbon Pricing Dashboard. Compliance Mechanisms: Coverage

⁴ Asia accounts for 36% of global gross domestic product (GDP) and 60% of the world’s CO₂ emissions as well as 80% of the world’s coal consumption (Asukawa 2021).

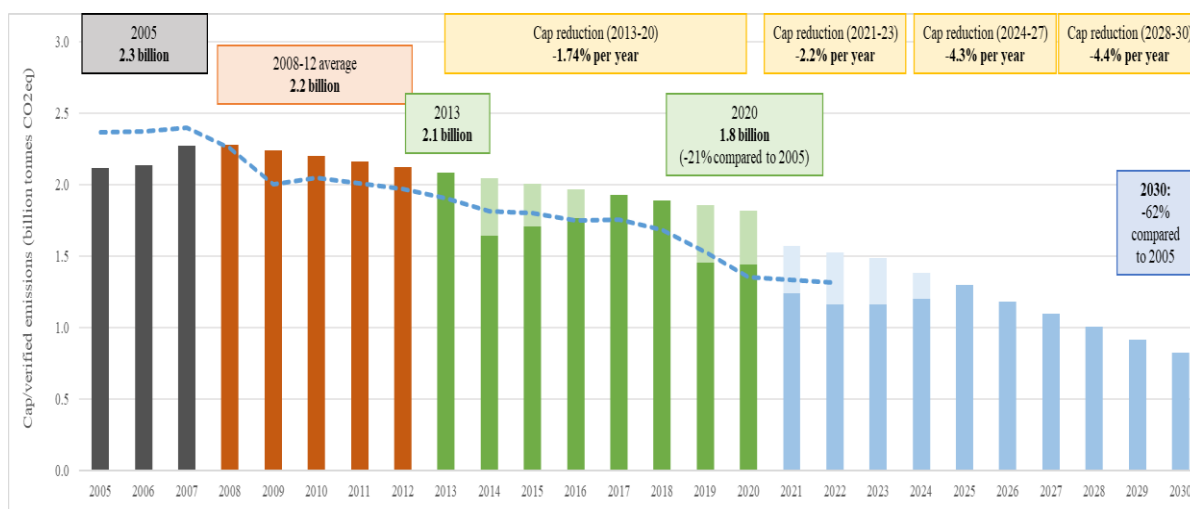
The paper is organized into five sections. Following the introduction (Section 1), the paper presents the case of the EU ETS, one of the oldest and largest systems, and the related empirical discussion on the same (Section 2). Section 3 examines the six ETSs currently operational in Asia and provides a comparative discussion of their characteristics. A framework to evaluate ETS policy success is presented in Section 4, followed by discussion of four key ETSs—i.e., those of the PRC, Kazakhstan, the ROK, and New Zealand. The evaluation discussion is limited to these since they have been in practice for some time.⁵ The paper ends with policy recommendations and lessons for countries going forward (Section 5).

II. THE EXAMPLE OF THE EUROPEAN UNION'S EMISSIONS TRADING SCHEME

Since 2005, the EU ETS has been a key instrument for decarbonizing the EU economy. It caps overall emissions of carbon dioxide (CO₂), nitrous oxide, and perfluorocarbons from over 11,000 heavy energy-using and electricity-generating installations (not companies) and aircraft, covering about 45% of the EU's GHG emissions. The cap is expressed in EU Allowances (EUA), where one allowance gives the right to emit 1 metric ton of carbon dioxide equivalent (CO₂e). For each year, companies must surrender enough allowances to account for all their emissions, otherwise heavy fines are imposed. The cap limits the total emissions allowed in the system and is set to bring emissions down by 62% by 2030 compared with 2005. Figure 1 illustrates the changes in the cap throughout all EU ETS phases, including the projected reduction until 2030.

⁵ Australia and Japan do not exactly have an ETS. Australia has a baseline-and-credit system (the Safeguard Mechanism), and that only since July 2023. Japan's Green Transformation ETS (GX-ETS) is a voluntary baseline-and-credit system that too became operational only recently, in October 2023. The Indonesian ETS (Nilai Ekonomi Karbon: NEK) started functioning in 2023 and has yet to produce sufficient data for an assessment.

**Figure 1: Emissions Cap Set in the European Union Emissions Trading Scheme
Compared with Verified Emissions**



CO₂e = carbon dioxide equivalent.

Legend: Bars = emissions cap, light shade bars in 2014–2016 = allowances backloaded in Phase 3, light shade bars since 2019 = feeds of allowances to the Market Stability Reserve, dashed line = verified emissions.

Source: European Commission (2023). [Report from the Commission to the European Parliament and the Council on the Functioning of the European Carbon Market in 2022 Pursuant to Articles 10\(5\) and 21\(2\) of Directive 2003/87/EC](#). Brussels. 31 October.

The EU ETS remains the world's most comprehensive cap-and-trade system for regulating GHG emissions at the installation level (despite the ETS of the PRC covering a bigger volume of emissions). The EU ETS is also the oldest GHG cap-and-trade system and the only multinational one, comprising 30 countries (27 EU Member States plus Iceland, Liechtenstein, and Norway). It is now in its fourth trading phase (2021–2030), following three phases—namely, a 3-year pilot (2005–2007), the EU's meeting of Kyoto Protocol commitments (2008–2012), and the EU's unilateral commitments to control GHGs (2013–2020). In the first two phases, permits (or allowances) were “grandparented”—i.e., allocated free to companies based on their past emissions. In the third and fourth phases, however, auctioning has become the main method of distributing allowances (at up to 57% of total), with free allocation, based on benchmarks, issued to reduce the risk of carbon leakage. Free allocation to specific sectors will be phased out between 2026 and 2034, in parallel with the phase-in of the EU Carbon Border Adjustment Mechanism for third-country imports of iron and steel, cement, aluminum, fertilizers, and hydrogen. Every year, covered entities must surrender allowances for their emissions under the EU ETS. The EU ETS was revised in 2023 in the context of the European Green Deal.⁶

⁶ Under the European Green Deal approved by the European Commission in 2020, all 27 EU Member States committed to turning the EU into the first climate-neutral continent by 2050. As an intermediate goal, these countries have also pledged to reduce GHG emissions by at least 55% by 2030 compared with 1990 levels.

The EU ETS has had to deal with several challenges. For example, monitoring, reporting, and verification (MRV) is carried out by individual Member States. As such, actual implementation of the EU ETS is only as good as countries' own capacity and willingness (Tol 2019). Second, a supply and demand mismatch for permits and price volatility continue to bedevil the market. In the first two phases (2005–2007 and 2008–2012), each country decided on the allocation of their emission allowances through national allocation plans (NAPs). This process established the EU-wide cap in a decentralized, bottom-up way (the sum of the NAPs was the overall cap). However, in the hope of creating a new export industry, countries allocated more permits than they should have (ibid.). Oversupply collapsed the market and carbon prices fell to zero and the European Commission took over the allocation of permits from Phase 3 (2013–2020) onward.⁷

To deal with the growing surplus of allowances, “backloading” (temporary withdrawal of emissions from the market) was implemented as a short-term measure in Phase 3. An auction of 900 million allowances set for 2014–2016 was postponed until 2019–2020. And, as a long-term measure, the Market Stability Reserve (MSR) was created in 2015 to place the growing surplus of allowances in the EU ETS. The MSR, which started operations in 2019, helped adjust auction volumes according to predefined thresholds for the total number of allowances in circulation. Based on the number in circulation, allowances are either withdrawn from auction and placed in the MSR or released from the MSR and auctioned to foster balance in the EU carbon market and strengthen resilience to demand shocks.

Despite measures to manage the surplus of allowances, EUA prices remained volatile (Figure 2), rising from about €20 per ton in January 2020 to a record high of €100.34 in February 2023, then falling back to about €62 in January 2024. Nonetheless, the way the MSR has worked suggests that concerns of “too low” and “ineffective” carbon prices in the early years of EU ETS may turn into concerns for too high prices (Gerlagh, Heijmans, and Rosendahl 2022).

⁷ This phase saw a major revamp of the EU ETS compared with the first two phases and brought about five key changes: (i) a single, EU-wide, cap on emissions in place of the previous system of national caps, (ii) auctioning as the default method for allocating allowances (instead of free allocation), (iii) harmonized allocation rules applying to the allowances still given away for free, (iv) more sectors and gases included, and (v) 300 million allowances set aside in the New Entrants Reserve to fund the deployment of innovative, renewable energy technologies and carbon capture and storage. See European Commission (nd).

**Figure 2: European Union Allowances—Historical Price Evolution, 2005–2024
(€/ton carbon dioxide equivalent)**



Source: Dimitrova, M. 2024. [EUA Price History: What Are the Drivers?](#) *Homaio*, 29 January.

The benefits of the EU ETS have been examined in the literature. Since its 2005 launch, the EU ETS has reportedly helped drive down emissions from electricity and heat generation and industrial production by 37.3%. It has also generated €184 billion (\$206 billion) in auction revenues for distribution to Member States since its third phase started in 2013, with €43.6 billion distributed in 2023 alone, rising from €31.1 billion in 2021 and €38.8 billion in 2022 (European Commission 2023). Overwhelmingly, Member States have used this revenue to invest in energy transformation and decarbonization and for social measures that support the green transition.

Growing evidence points to significant impacts of the EU ETS, based on country-specific microdata studies (Petrick and Wagner 2014; Klemetsen, et al. 2020; Colmer, et al. 2022) and data from different sectors (Bayer and Aklin 2020). A recent cross-country study of installation data in France, the Netherlands, Norway, and the UK found its introduction was associated with a statistically significant 10% reduction of carbon emissions in the first two trading phases between 2005 and 2012 (Dechezleprêtre, Nachtigall, and Venmans 2023). Data collected from firms in all ETS-regulated countries show the EU ETS has had no significant effect on profits and employment and has led to an increase in regulated firms' revenues and fixed assets (ibid.).

III. ASIAN NATIONAL EMISSIONS TRADING SCHEMES: A COMPARATIVE DISCUSSION

Out of 14 national ETSs worldwide, 7 are in Asia.⁸ National ETSs of various types are in operation in Australia, the PRC, Indonesia, Kazakhstan, the ROK, and New Zealand. NZ ETS dates to 2008 and is the second oldest in the world.⁹ NZ ETS was followed by Kazakhstan's in 2013 and the ROK's in 2015. Between 2013 and 2016, the PRC initiated six subnational ETSs, which set the pathway for a nationwide network of ETS in 2021. Australia does not have a formal ETS but adopted a mandatory baseline-and-credit system in 2023.¹⁰ Indonesia is the most recent entrant, with its ETS starting in 2023, while in Japan a national ETS is still under development (being preceded by the two subnational ETSs, in Tokyo and Saitama).¹¹ This section looks at the six national ETSs and excludes that of Japan, which is still under development.

Contextually, the six nations with ETSs (plus Japan) differ from each other (Table 1). In absolute numbers, the PRC has the largest population and economy and New Zealand the smallest. While Australia is the richest on a per capita income basis, it has the highest per capita CO₂ emissions, reflecting its significant economic activities in mining and resource-based industries. Indonesia features modestly in both per capita gross domestic product (GDP) and CO₂ emissions. Of the six countries, New Zealand leads in terms of its share of electricity from renewables, with Kazakhstan and the ROK at the other extreme. Between 1990 and 2022, total GHG emissions increased dramatically in the PRC, Indonesia, and the ROK but declined in Japan, Kazakhstan, and New Zealand while remaining about the same in Australia. For all ETS countries except for New Zealand, energy generates the highest share of emissions. For the other sectors, industry constitutes the highest share for the PRC, agriculture for Australia, Kazakhstan, and New Zealand, and waste for Indonesia.

⁸ The 14 national ETSs are Australia, Austria, Canada, the PRC, Germany, Indonesia, Japan (under development), Kazakhstan, the ROK, Mexico, Montenegro, New Zealand, Switzerland, and the UK.

⁹ The first ETS for GHGs was the EU ETS, established in 2005.

¹⁰ Unlike an ETS, a baseline-and-credit system imposes no emission caps. Firms below a given baseline can generate and sell credits to other firms that exceed the baseline.

¹¹ The Tokyo Cap-and-Trade Program (2010) and the Saitama Target Setting Emissions Trading System (2011).

Table 1: Comparative Economic and Environmental Data for Countries in Asia and the Pacific with Emissions Trading Schemes and the European Union

Economy	1	2	3	4	5	6	7	8	9	Overall GHG Emissions by CRF Category Excluding LULUCF* (%)			
	Population millions (2022)	GDP in billion, constant 2015 \$ (2022)	Per capita GDP, constant 2015 \$ (2022)	CO ₂ , metric tons per capita (2020)	% of electricity from renewables (excluding hydro) (2015)	% of forest area (2021)	Net zero by	GHG emissions, million tons of CO ₂ e (1990)	GHG emissions, million tons of CO ₂ e (2022)	Energy	Industry	Agriculture	Waste
Australia	26.0	1,586.0	60,994	14.8	8.3	17.4	2050	536.9	573.6	79	7	12	2
PRC	1,410.0	16,325.0	11,560	7.8	4.9	23.6	2060	4,407.6	13,943.7	78	15	6	1
Indonesia	275.5	1,122.0	4,074	2.1	4.8	48.4	2060	1,184.3	2,127.2	67	7	12	14
Japan	125.1	4,529.0	36,203	8.0	7.8	68.4	2050	1,222.8	1,094.2	87	9	3	1
Kazakhstan	19.6	221.6	11,291	11.3	0.2	1.3	2060	419.5	357.96	78	9	12	1
ROK	51.6	1,740.9	33,719	11.0	1.5	64.3	2050	292.2	646.1	87	8	3	2
New Zealand	5.1	218.1	42,559	6.2	24.6	37.6	2050	76.2	65.4	40	5	51	4
EU (27 nations)	447.4	15,281.7	37,433	5.5	18	39.9	2050	4,806.8	3,343.1	76	10	11	3

CO₂ = carbon dioxide, CO₂e = carbon dioxide equivalent, CRF = Common Reporting Format, EU = European Union, GDP = gross domestic product, GHG = greenhouse gas, LULUCF = land use, land use change, and forestry, PRC = People's Republic of China, ROK = Republic of Korea.

*LULUCF CRF is the standardized way for Parties to the United Nations Framework Convention on Climate Change to report emissions inventories (see UNFCCC nd).

Sources:

Column 1–6: [World Bank data](#) (accessed 6 April 2024).

Column 8 and 9: Jones, M., et al. 2024. [National Contributions to Climate Change Due to Historical Emissions of Carbon Dioxide, Methane and Nitrous Oxide](#). Scientific Data. Zenodo, 19 March, with major processing via Ritchie, H., P. Rosado, and M. Roser. 2020. [Greenhouse Gas Emissions](#). Our World in Data (accessed 6 April 2024).

Column 10: International Carbon Action Partnership [ETS Map](#) (accessed 6 April 2024).

The same set of countries share the objective of meeting their respective GHG mitigation commitments. These commitments entail the aim to achieve carbon neutrality by 2050 for Australia, Japan, the ROK, and New Zealand and by 2060 for the PRC, Indonesia, and Kazakhstan. All also intend to make significant GHG reductions by 2030. This is noteworthy because the PRC is the world's largest GHG emitter, while Indonesia is 6th, Japan 7th, the ROK 13th, and Australia 15th (EDGAR 2024).

However, the countries' ETS systems vary in their ambition, design, and implementation (Table 2). In terms of scope and GHG coverage, the ROK ETS (K-ETS) is the most comprehensive, since it includes all the Kyoto Protocol gases. NZ ETS is also comprehensive, though it excludes biogenic methane, which is a major emission category for the country. Most of the other ETSs focus on CO₂ emissions. They cover a share of total emissions that ranges from a low of 26%–29% (Australia and Indonesia) to a high of 88.5% (ROK). K-ETS and the NZ ETS are the most comprehensive; the others focus mostly on the power sector, a major source of CO₂ emissions.

Table 2: Comparison of the Main Features of Emissions Trading Schemes

Features	Australia	PRC	Indonesia	Japan	Kazakhstan	ROK	New Zealand
1. GHG reduction targets for the country	By 2030: 43% below 2005 levels By 2050: Net zero emissions	By 2025: 18% below 2020 levels By 2030: 65% below 2005 levels using a “1+N” policy framework By 2060: carbon neutrality	By 2030: 31.9% below Business As Usual including LULUCF By 2060: Climate neutrality	By 2030: 46% below from 2013 levels including LULUCF By 2050: Net zero GHG emissions	By 2030: 15% (unconditional) to 25% (conditional) below 1990 levels of GHGs By 2050: 40% reduction in power sector emissions from 2012 levels By 2060: Carbon neutrality	By 2030: At least a 35% reduction below 2018 emissions By 2050: carbon neutrality	By 2030: 50% reduction of net emissions below gross 2005 levels By 2050: Reduce net emissions of all GHGs (except biogenic methane) to zero
2. Year ETS introduced	2016	2021	2023	Being developed	2013	2015	2008
3. GHGs covered	CO ₂ , CH ₄ , N ₂ O, SF ₆ , HFCs, PFCs	CO ₂	CO ₂ , CH ₄ , N ₂ O,	CO ₂	CO ₂	CO ₂ , CH ₄ , N ₂ O, SF ₆ , HFCs, PFCs	CO ₂ , CH ₄ , N ₂ O, SF ₆ , HFCs, PFCs (but no biogenic CH ₄)
4. Covered emissions (% total emissions)	26% (year 2020)	40% (year 2021)	29% (year 2023)		47% (year 2021)	88.5% (year 2021)	48% (year 2021)
5. Emissions subject to ETS (MtCO ₂ e)	137 MtCO ₂ e	5,200 MtCO ₂ e (approx.)			160 MtCO ₂ e	599 MtCO ₂ e	36.9 MtCO ₂ e
6. Sectoral coverage	Waste, domestic aviation, transport, industry	Power	Power		Industry, power	Waste, domestic aviation, transport, buildings, industry, power	Forestry, maritime, waste, domestic aviation, transport, buildings, industry, power
7. No. of obligated entities/firms	215 (year 2023)	2257 (2021–2022)	In 2023: 42 entities (99 installations) In 2024: 63 entities (146 installations)	570 companies	128 companies (201 installations)	804 (year 2023)	4,114 (year 2023) (Only 246 have mandatory obligations, the rest are voluntary)
8. Point of regulation	Point source	Point source	Point source	Point source	Point source	Point source (all sectors except buildings regulated downstream) ^a	Upstream (power, aviation, buildings, forestry, transport); point source (industry, waste)
9. Development phases	No phases: current baseline decline rate of facilities has been fixed till 2030	First cycle (trading-cum-compliance) for 1 year (2021), for emissions in 2019/2020	Phase I: 2023–2024 Phase II: 2025–2027 Phase III: 2028–2030	Phase I: 2023–2026 (voluntary baseline-and-credit system)	Phase I: 2013 Phase II: 2014–2015 2016–2017: ETS suspended Phase III: 2018–2020	Phase I: 2015–2017 Phase II: 2018–2020 Phase III: 2021–2025	No specific phases

Features	Australia	PRC	Indonesia	Japan	Kazakhstan	ROK	New Zealand
		Second cycle for 2 years (2022 and 2023), for emissions in 2021/2022		Phase II: 2026 onward (mandatory ETS with auctioning from 2033)	Phase IV: 2021 Phase V: 2022–2025		
10. Type of cap	Baseline-and-credit	Bottom-up (benchmarking using emissions intensity)		Voluntary baseline-and-credit	Bottom up (grandparenting + benchmarking)	Absolute (top down)	
11. Cap/total emission limit	Net emissions should not exceed 100 MtCO ₂ e in 2030 and should reach zero by 2050	About 4,500 MtCO ₂ in 2019 and 2020 and about 5,000 MtCO ₂ in 2021 and 2022	Coal-fired power plants (25–100 MW): 1.3 tCO ₂ e /MWh Mine mouth plants (≥ 100 MW): 1.1 tCO ₂ e /MWh Non-mine mouth plants (100–400 MW): 1.0 tCO ₂ e /MWh Non-mining plants (≥ 400 MW): 0.9 tCO ₂ e/MWh		649.8 ^b MtCO ₂ for overall period, with declining annual caps in the latest phase	3048.3 ^c MtCO ₂ (plus New Entrants' Reserve of 610 MtCO ₂) in last phase	2024: 27.9 MtCO ₂ e
12. Allowance allocation	Set up based on production levels multiplied by an emission intensity value	Output-based benchmarking for 4 benchmarks: conventional coal plants below 300 MW; above 300 MW; unconventional coal and natural gas Free allocation currently	Auctioning and benchmarking		Free allocation (benchmarking and grandparenting approach)	Free allocation (benchmarking and grandparenting approach) and auctioning	Free allowance and auction Note: Proportion of allowances offered at auction in 2023 was 54% but none were sold (winning bid less than reserve price)
13. Compliance period	Annual	Annual	Annual		Annual	Annual	Annual
14. Banking and borrowing	Unlimited banking of Safeguard Mechanism Credits is allowed up to 2030. Borrowing up to 10% of a facility's baseline each year will be allowed until 2030, with a 10% interest rate applied in the year after borrowing occurs.	Borrowing was not allowed in 2019 to 2020. In the 2021 to 2022 allocation plan, borrowing is allowed. Companies with a shortfall of 10% or more can apply to borrow from a pre-approved allocation for 2023, up to 50% of the shortfall.	Banking is allowed within phases, though allowances (PTBAE-PU) are valid for a maximum of 2 years from the end of the previous compliance period. Banking is not allowed across phases. Borrowing is not allowed.		Banking is allowed within each phase, but not between phases. Borrowing is not allowed.	Banking is allowed with restrictions across and within phases. Borrowing is allowed within a single trading phase. For the latest phase, Phase 3, in the first trading year entities could borrow up to 15% of their compliance	Banking is allowed. Borrowing is not allowed.

Features	Australia	PRC	Indonesia	Japan	Kazakhstan	ROK	New Zealand
	For the first 2 years, this interest rate will be lower, at 2%. The facility's baseline would then decrease by a corresponding amount the following year, with, in addition, 2% interest for the 2024/25 and 2025/26 financial years, and 10% interest for 2026/27.	Banking from 2019 to 2020 was allowed in 2021 to 2022. Future rules on banking are not yet defined				obligation. From the second to fourth trading years, the same borrowing formula as for 2019 applies.	
15. Average prices (2023)		Secondary market: CNY 68.35 (\$9.65)	Auction: IDR 10,000 (\$0.64) Secondary market: IDR 69,600 (\$4.45)		Secondary market: KZT 473 (\$1.04)	Auction: KRW 10,672 (\$8.17) Secondary market: KRW 9,999 (\$7.66)	Auction: NZD 0 (\$0) Secondary market: NZD 62.79 (\$38.30)
16. Total revenue						\$901 million (cumulative) \$64.4 million (2023)	\$3.1 billion (cumulative) \$21.3 million (2023)
17. MRV	Monitoring: Facilities need to apply for a site-specific emissions intensity value for the first compliance year. Reporting: Annual self-reporting is used, according to the guidelines defined in the National Greenhouse and Energy Reporting Act. Verification: All facilities with emissions greater than 1 MtCO ₂ e per year are required to undergo an independent emissions audit each year.	Monitoring: Covered entities are required to set up monitoring plans and monitor their emissions based on these. Reporting: Covered entities must submit the previous year's emissions reports by the end of April each year. Verification: Provincial-level ecological and environmental authorities are responsible for organizing the verification of GHG reports. Verification of emissions from the power sector must be complete by the end of June.	Monitoring: An MRV system is in operation in the industry sector and the power generation subsector. Pilot MRV programs are being conducted in the cement and fertilizer sectors. Reporting: Reports must be submitted by the end of January for the preceding year. Installations must report emissions of considered GHGs expressed in units of CO ₂ e. Verification should be completed by the end of March, following the January reporting deadline. Verifiers are required to follow the		Reporting is required annually for installations above the 20,000 tCO ₂ e/year threshold. Verification: Emissions data reports and their underlying data require third-party verification by an accredited auditor.	Reporting: Annual reporting of emissions from the previous year must be submitted by the end of March. Verification: Emissions must be verified by a third-party verifier. Framework: Emissions reports are reviewed and certified by the Certification Committee of the Ministry of Environment by the end of May.	Reporting: Most sectors are required to report annually; the deadline is the end of March to submit an annual emission return. Verification: This follows a system of self-reporting supplemented by a program of official government audits. Third-party verification is not required for emissions reports. However, participants must seek third-party verification if they apply for the use of a unique emissions factor, as opposed to using the default factors supplied by the government.

Features	Australia	PRC	Indonesia	Japan	Kazakhstan	ROK	New Zealand
			guidelines for GHG emission verification in the power subsector.				
18. Enforcement	The maximum civil penalty is set at 1 penalty unit per tonne of excess emissions per year. As of January 2023, a penalty unit is AUD 275 (\$182). The infringement notice is charged at one-third of the maximum civil penalty to a maximum of 150,000 penalty units.	According to Interim Regulation, fines for not submitting reports will increase from CNY 10,000–30,000 to CNY 50,000–200,000 (\$7,058–28,232), while fines for failing to comply will increase from CNY 20,000–30,000 to 5 to 10 times the market value of missing allowances (based on the average price in the month before the compliance deadline).	Entities failing to report their emissions or participate in carbon trading by the end of the period will see a 25% reduction in their PTBAE-PU.		The noncompliance penalty equals 5 monthly standard units for each tCO ₂ . This was KZT 18,460 (\$40.47) in 2024.	The penalty shall not exceed either 3 times the average market price of allowances of the given compliance year or KRW 100,000 (\$76.58) per tone.	An entity that fails to submit an emissions report by the due date and to surrender or repay emissions units must pay a fine.

AUD = Australian dollar, CNY = Chinese yuan, ETS = emissions trading scheme, GHG = greenhouse gas, HFC = hydrofluorocarbon, IDR = Indonesian rupiah, KRW = Republic of Korea won, KZT = Kazakhstani tenge, LULUCF = land use, land use change, and forestry, MRV = monitoring, verification, and reporting, MtCO₂e = metric ton of carbon dioxide equivalent, MW = megawatt, NZD = New Zealand dollar, PFC = perfluorocarbon, PRC = People's Republic of China, PTBAE-PU (Persetujuan Teknis Batas Atas Emisi-Pelaku Usaha, Indonesia's allowances), ROK = Republic of Korea, SF₆ = sulfur hexafluoride.

^a Downstream regulation of end users in the supply chain such as consumers.

^b Caps for each phase: 2022: 166.2 MtCO₂ (plus a reserve of 11.8 MtCO₂); 2023: 163.7 MtCO₂ (plus a reserve of 11.6 MtCO₂); 2024: 161.2 MtCO₂ (plus a reserve of 11.5 MtCO₂); 2025: 158.8 MtCO₂ (plus a reserve of 11.3 MtCO₂).

^c Caps for each phase: 2021: 589.3 MtCO₂e; 2022: 589.3 MtCO₂e; 2023: 589.3 MtCO₂e; 2024: 567.1 MtCO₂e; 2025: 567.1 MtCO₂e.

Source: International Carbon Action Partnership [ETS Map](#) (accessed 6 April 2024).

ETS design also varies by country, with only NZ ETS and K-ETS as top-down absolute caps.¹² Others take a more bottom-up approach, whereby the cap depends on firms' output levels multiplied by a benchmark emissions intensity (in other words, it is endogenous). Of particular interest is the cap determination in the PRC's ETS. This not only uses a bottom-up approach but also allocates allowances in two stages during each compliance period. At the beginning of the compliance period, a firm receives allowances, a_{t-1} , equal to its output from the previous compliance period, q_{t-1} , multiplied by the sector's benchmark emissions–output ratio, $\bar{e}$, and an 'initial allocation factor', α .¹³ Toward the end of the compliance period, the firm receives additional allowances, a_t , needed to bring the ratio of total allowances to end-of-period output (q_t) in conformity with the sector-specific benchmark emissions-output ratio, $\bar{e}$ (Goulder, et al. 2017).¹⁴ In effect, the PRC ETS is a tradable performance standard (TPS) that implicitly subsidizes electricity output (Goulder, et al. 2022).

Allowances are allocated through grandparenting and/or benchmarking.¹⁵ Another interesting case is the Kazakhstan ETS (KazETS), which has gone through few iterations in design—from only grandparenting (Phases I and II), to a dual allocation system (Phase III) where firms could opt for grandparenting or benchmarking, to only benchmarking (Phases IV and V). Benchmarks for coal-generated energy are lax, in an attempt to accommodate coal-fired plants with outdated technology. However, this “weakens any incentives rooted in this trading system for a transition from to gas-fired power” (Kazenergy 2019, Section 4.4.2, p. 101). There is very little auctioning in any ETS. In NZ ETS, 54% of allowances in 2023 were offered for auction but none were sold since the clearing price was lower than the reserve price.

There is very little auctioning in any ETS, and allowances are mostly given for free, especially for firms that are emissions-intensive-trade-exposed (EITE). As we have seen, in NZ ETS, 54% of allowances were offered for auction in 2023 but none were sold. NZ ETS and K-ETS are the only two with a long enough time series of prices for an effective allowances market.¹⁶ Compared with in the EU ETS, allowance prices have been low in K-ETS and moderate in NZ ETS (Figure 3). Also, in line with the experience of the EU ETS, allowance prices in both

¹² The New Zealand ETS was originally designed to operate without a specific domestic cap, as this accommodated (any amount of) carbon sequestration from forestry activities and a full link to the international Kyoto Protocol carbon markets. This ended in 2015.

¹³ Four sector-specific benchmarks are defined, for (i) coal plants below 300 MW, (ii) coal plants above 300 MW, (iii) unconventional coal plants, and (iv) natural gas plants.

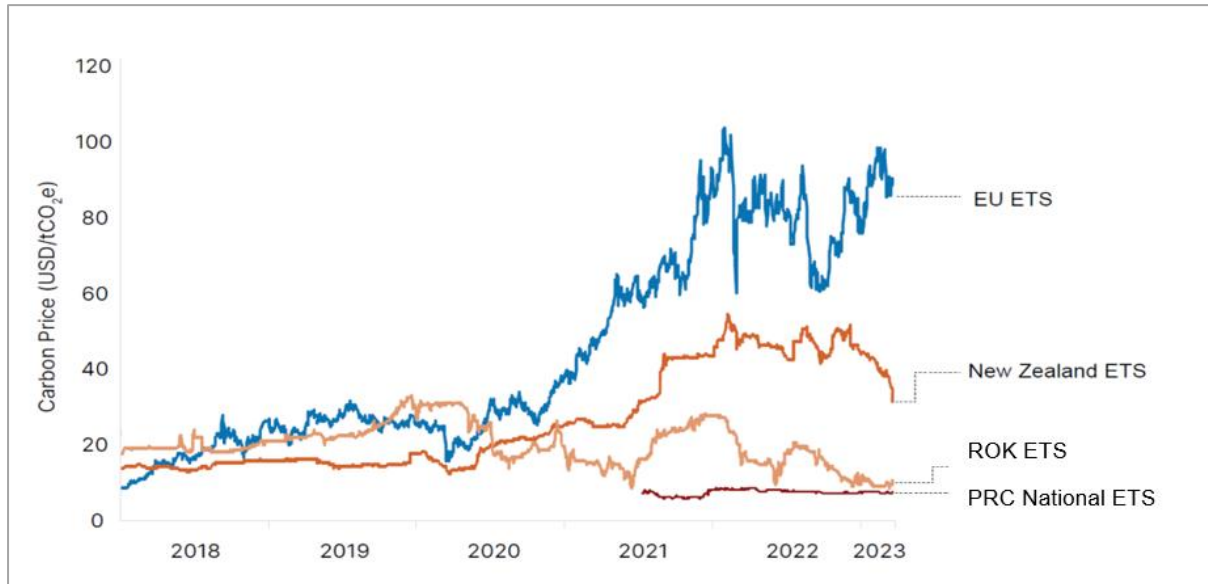
¹⁴ The quantity of additional allowances (a_t) is given by the expression $a_t = \bar{e} \cdot q_t - \alpha \cdot \bar{e} \cdot q_{t-1}$. The first term, on the righthand side, is the total number of end-of-period allowances that would be consistent with the emissions intensity benchmark $\bar{e}$. The second term is the number of allowances given to the firm at the beginning of the compliance period. Thus, the difference between the two is the number of additional allowances needed.

¹⁵ For a discussion of allocation methods, see ICAP (nd).

¹⁶ The ETS market in the PRC is in its infancy but data for 2 years is shown in the table. KazETS has had stops and starts, and allowance prices are not significantly different from zero (Table 2, Row 15).

markets have been volatile. Prices in New Zealand fell to near zero (US\$2) in 2013 soon after NZ ETS was delinked from Kyoto markets but have been on an upward trend since.

Figure 3: Price Evolution in Selected ETSs from 2018 to 2023



ETS = emissions trading scheme, PRC = People's Republic of China, ROK = Republic of Korea, tCO₂e = ton of carbon dioxide equivalent.

Source: World Bank. 2023. State and Trends of Carbon Pricing 2023 (Figure 1).

Several ETSs have been implemented over multiple phases, starting with limited rollout. KazETS has had as many as five phases (and between phases was suspended for 2 years) and K-ETS and Indonesia's Nilai Ekonomi Karbon (NEK) over three phases. The PRC's ETS has two compliance periods. The Australian Safeguard Mechanism has no phases and the current rate of decline of emissions is fixed until 2030. NZ ETS also has no phases. Indonesia has announced three phases, with the current phase covering 2023–2024.

The national ETSs are significantly different from a theoretical ETS that would be considered "first best." One should note that a key objective of an ETS is to achieve a given reduction in emissions at least cost. It does so by pricing emissions and by equating the marginal abatement cost (MAC) across all sources. Hence, a "first best" ETS would comprise the following features:

- There is an absolute cap on emissions. This cap is exogenously determined. In other words, it is top-down (and typically becomes tighter over time to meet environmental goals).

- The method of initial allocation of allowances—free allocation (usually through grandparenting) or auction (or a combination of the two)—does not affect the cost-effectiveness of the ETS.¹⁷
- All sources of emissions are covered to achieve the full cost savings potential of equating MACs.¹⁸
- There is a perfect market for emissions with many buyers and sellers. In other words, there is no market power and/or “thin” markets.
- The market clears in each trading period (usually a year)—i.e., demand and supply are balanced. There is no banking and borrowing. Nor does the regulator manipulate permit prices by injecting/withdrawing liquidity.
- Prices of the goods whose production generates emissions (e.g., electricity) are not regulated. If emissions are restricted, marginal costs of producing the good will increase. If prices are not allowed to increase the market will not clear.¹⁹
- Other environmental regulations (such as command and control) do not operate simultaneously.

None of the six national ETSs resemble a theoretical, or “first best,” ETS. They instead reflect an approach to make them acceptable to stakeholders and to accommodate other “real world” contexts. Some of the points of departure, also discussed earlier, are:

- The cap is determined bottom-up and depends on a firm’s output, particularly in the case of the PRC. It therefore is endogenous and not predetermined. In Kazakhstan, the cap is also determined bottom-up through a mix of grandparenting,²⁰ benchmarking, and auctioning. In fact, each firm can choose between grandparenting and benchmarking .

¹⁷ It is only a question of who appropriates the scarcity rent (of permits). With auctions, it is the government; under grandparenting, the rent accrues to firms. However, this makes a big difference to how acceptable stakeholders (firms) consider an ETS to be. Free allocation can reduce opposition to ETS; another way to achieve this is by recycling auction revenue (but this is more indirect).

¹⁸ Equating MAC across sources in any given period is akin to *static efficiency*. By contrast, *dynamic efficiency* would mean that creating a price for emissions provides an incentive (a signal) for investment in research and development, new technology, and so forth, lowering the MAC over time (Milliman and Prince 1989, 1992).

¹⁹ Note that the price of good will increase irrespective of whether permits are auctioned or given free. With auctions, the scarcity rents accrue to the government; when free, scarcity rents accrue to the firms (Phaneuf and Requate 2017, Figure 5.4). Also see footnote 17 above. The *extent of price increase* will depend on the elasticity of demand, which determines how much of the increase in cost can be passed (ibid., Figure 5.3).

²⁰ This grandparenting is not to be confused with that under an exogenously determined top-down cap. In that case, grandparenting is done *after* a cap is decided and applies to *all sources*. In a bottom-up system, however, grandparenting is decided by the installation (source). For example, in Phase 3 of KazETS, installations were given the choice between grandparenting and benchmarking.

- There are various exemptions, especially for EITE firms, to which allowances are given free and can cover as much as 100% of their emissions (e.g., in K-ETS).²¹
- If permits are auctioned, a reserve price may be set and the auction fails if no one bids above it (as observed in the case of New Zealand).
- Many times, major sources of emissions are not covered. In New Zealand, agriculture contributes more than half of the country's GHG emissions but does not come under ETS regulations.²²
- There is extensive use of banking and (at times) borrowing, for example in NZ ETS and KazETS.
- Governments sometimes intervene in the allowance market to manage liquidity (to manage prices). The ROK formally uses "market makers" (third-party participants) in the ETS who can draw on a separate government-held reserve of allowances to increase liquidity.
- Product markets are regulated (in Kazakhstan), which means higher costs that result from restrictions on emissions cannot be passed on and the market is not in equilibrium.

IV. EVALUATING ASIA AND THE PACIFIC EMISSIONS TRADING SCHEMES

Although the ETSs are still evolving, so far some are more successful than others. This section adopts a framework to discuss ongoing initiatives around the ETS of four countries—the PRC, Kazakhstan, the ROK, and New Zealand. It does not discuss Australia and Japan as they do not have an ETS per se. Australia has a baseline-and-credit system (the Safeguard Mechanism), operational since July 2023. The Japan Green Transformation ETS (GX-ETS) is a voluntary baseline-and-credit system that became operational only recently, in October 2023. Since the Indonesian ETS (NEK) started in 2023, there is not enough data as of now for an assessment.

A. The Assessment Framework

A successful ETS requires a sufficiently tight emissions constraint coupled with effective MRV to ensure emissions are reported accurately and the cap is being enforced. There should also be confidence in the level of long-term prices to drive investment in low-carbon solutions. Minimizing the risk of carbon leakage is another determinant of environmental effectiveness,

²¹ EITE firms are in sectors that are polluting and exposed to international trade. An ETS puts them at a competitive disadvantage compared with in countries that do not regulate emissions.

²² A proposal to impose a tax on biogenic methane emissions (from sheep and cattle) was shelved after a change in government in 2023.

as is ensuring the integrity of emission units, such as offset credits entering the system from outside the cap. The World Bank (2021) proposes a set of criteria to determine the design and implementation success of an ETS, which is adapted in this paper as an assessment framework.²³ The eight criteria are its contribution to mitigation; cost-effectiveness; predictability; robustness; accountability and transparency; administrative cost-effectiveness; fairness; and how well it fits national circumstances and is compatible with other GHG mitigation policies. Table 3 summarizes the eight criteria, with related indicators. Appendix 2 describes each criterion in more detail.

The same framework is used in the next section to qualitatively assess the ETSs of the PRC, Kazakhstan, the ROK, and New Zealand.

Table 3: Assessment Framework for Comparing Emissions Trading Schemes

Criterion	Definitions	Indicators
<i>Contribution to mitigation</i>	Degree to which achieved outcomes correspond to intended goals	1. Scope (% of emissions covered, coverage of key emitting sectors, any important sector omitted?) 2. Stringency of emissions cap 3. Emissions reductions attributable to ETS
Cost-effectiveness of mitigation	Achieving emissions target using cheapest means possible irrespective of which GHG is reduced, in which sector, or at which point in time	1. Liquidity of ETS market 2. Permit prices 3. EITE sector 4. Level of energy price and cost pass-through regulations
Predictability	Ability for regulated firms as well as others to prepare and consider ETS and its implications	1. Abrupt changes in ETS operation, rules, etc. 2. Allocating allowances to new entrants and removing allowances for closures 3. Banking and borrowing. 4. Longer commitment periods
Robustness	Ability to cope with time inconsistency and provide carbon price predictability and credibility over time	1. Use of price or supply adjustment measures (PSAMs) 2. Ability to integrate new participants 3. Use of offsets as compliance flexibility mechanism
Accountability and transparency	Level of enforcement and compliance instruments available as well as meaningful access to information	1. Compliance with international MRV standards 2. Severity of penalties for breaches of accountability 3. Independent body oversees allocation and assessment methods 4. Availability and third-party monitoring of data 5. Operating according to well-developed policies
Administrative cost-effectiveness	Evaluating administrative capacity to operate an effective system, which includes ability to operate	1. Length of compliance period 2. Transaction costs—e.g., for brokerage services

²³ Our work borrows from and adapts Howie, et al. (2020), who based their assessment criteria on an earlier version of the Partnership for Market Readiness–ICAP–World Bank Handbook (first edition) (World Bank 2016). Some criteria such as robustness were added in the second edition (World Bank 2021), which we use.

Criterion	Definitions	Indicators
	an MRV system to international standards at low administrative cost	3. Design of ETS (upstream or downstream) and participation thresholds 4. Effective MRV and administration
Fairness	Evaluating regressive impacts and procedural fairness of the ETS well as redistribution of any revenue	1. Types and shares of different allocation methods 2. How well allocation methodologies are harmonized 3. Period for averaging historical emissions if grandparenting is used 4. Programs for low-income consumers 5. Vertical equity characteristics of contribution to GHG reduction
Tailoring to national circumstances and compatibility with other policies for GHG mitigation and more generally	Managing policy interactions Political economy of ETS or of GHG mitigation more generally (e.g., Australia, New Zealand)	1. Role of ETS in climate policy mix 2. Impact of companion policies on ETS outcomes 3. Impact of ETS on the attainment of companion policy objectives

EITE = energy-intensive and trade-exposed, ETS = emissions trading scheme, GHG = greenhouse gas, MRV = monitoring, reporting, and verification, PSAM = price or supply adjustment measure.

Source: Adapted from World Bank. 2021. [Emissions Trading in Practice: A Handbook on Design and Implementation](#). Second edition.

B. Assessment of Four ETSSs

We apply the criteria discussed above and their indicators (Table 3) to assess the four featured ETSSs, with NZ ETS first, followed by KazETS, K-ETS), and then the PRC's ETS.

Contribution to Mitigation

NZ ETS covers a wide range of sectors and GHGs (though it falls short of its original goal of “all sectors all gases”). The emission cap declined by about 15% between 2021 and 2024—from 32.8 MtCO₂e to 27.9 MtCO₂e (World Bank 2021). Of the allowances offered for auction in 2023, 54% remained unsold and were retired, implying a tightening of supply in the market. Post-2020 reforms and rising allowance prices have boosted the forestry sector by incentivizing afforestation (Carver, et al. 2022).

KazETS is limited to the industry and power sectors and covers 128 companies. Its record on design and implementation is mixed. Few transactions have been made under the system, owing to generous benchmarks, a high number of free quotas, and a low carbon price (IEA 2022). The cap needs to be lowered and benchmarks tightened. There have also been starts and stops—in 2026–2017 the ETS was abruptly suspended and then restarted.

K-ETS has contributed significantly to mitigating GHG emissions (covering about 89% of GHGs) in the ROK. It covers a wide range of sectors²⁴ and has reduced emissions in industries like power generation and manufacturing. Its sector coverage and gradual reduction of emission caps align with the country's climate goals, although deeper emissions cuts may be needed to meet long-term targets.

The PRC's ETS is in the early stages of implementation. It is limited to the power sector that is regulated and covers only CO₂. The emissions cap is not absolute (i.e., top-down). There is room for the ETS to tighten the cap²⁵ to achieve GHG reduction targets.

Cost-Effectiveness of Mitigation

NZ ETS covers several sectors and gases and so can take advantage of differences in abatement across sources and sectors—especially of CO₂ removals from forestry. Liquidity in carbon trading is increased by the operation of a secondary market.²⁶ However, auction failures, where bids are below the reserve price, can tighten market supply. Free allocation is being phased down and is granted only for EITE activities (6.1 million New Zealand Units, or NZUs, were allocated for industrial EITE activities in 2022). Over time, allowance prices in secondary market have been rising, and they have been substantive in recent years (about \$40–\$50/ton).²⁷

KazETS has limited scope to equate abatement costs across sources given that most sources are state-owned entities, the market is illiquid with very little trading, and allowance prices are not significantly different from zero (about \$1/ton).

K-ETS aims to achieve emissions reductions at a reasonable cost by employing market-based mechanisms such as auctions, banking, and offsets. During Phase 2, market makers were introduced (third-party participants) to inject liquidity (i.e., allowances) into the market. There were eight K-ETS market makers at the start of 2024. To boost liquidity, these institutions may access the government's reserve of 20 million allowances.²⁸ There is no differentiated rule of allowance price for EITE. The rule also includes indirect emissions for large electricity

²⁴ Waste, domestic aviation, transport, buildings, industry, and power.

²⁵ The cap is estimated to have increased by about 11.11% between 2020 and 2022—from 4,500 MtCO₂ to 5,000 MtCO₂.

²⁶ Businesses participating in NZ ETS can buy and sell New Zealand Units (NZUs) from each other in the secondary market. The secondary market is an important part of NZ ETS because it ensures trading gives companies a strong incentive to save money by cutting emissions in the most cost-effective way (MfE n.d.).

²⁷ See [ICAP Allowance Price Explorer](#).

²⁸ [ICAP factsheet on K-ETS](#) (accessed 25 June 2024).

consumers. Stringent price ceiling policy regulations of the electricity generation sector forbid CO₂ costs to be passed on to consumers' electricity prices.

The PRC's ETS is not an ETS but a TPS, and has other objectives (equity across regions) than cost-effectiveness (Goulder, et al. 2017, 2021).

Predictability

NZ ETS provides a stable and predictable framework for long-term emissions reduction. It is governed by the Climate Change Response Act of 2002.²⁹ The Act outlines the scheme's objectives, design, and rules, ensuring participant understanding and long-term commitment. Predetermined annual emissions caps, aligned with long-term targets like net zero by 2050, provide clarity for planning and investment. NZ ETS allows allowances to be banked, which can enhance long-term planning, investment certainty, and market stability. It can also encourage early emissions reductions, contributing positively to predictability. The NZ ETS prohibition on borrowing limits short-term flexibility but strengthens long-term predictability by ensuring that emissions reductions are not deferred and that each compliance period's cap is strictly met. This maintains the integrity of the ETS and ensures a steady approach to achieving emissions targets.

KazETS has gone through several phases, with changes in allocation methods, one-off free allowances, temporary suspension of the ETS, and so forth. For Phase 3, new entrants receive allowances based on sector (and fuel type if available) benchmarks. There have also been different commitment periods, which may create uncertainties. For instance, Phase 1 had a 1-year commitment period; Phase 2 had a 2-year commitment period; and Phase 3 had a 3-year commitment period. Banking is allowed within each phase but not between phases. Borrowing is not allowed. Prohibiting banking between phases can lead to significant price fluctuations at the end of each phase. Allowances may become extremely cheap if firms have surplus allowances they cannot carry over. They may become very expensive if firms need to purchase allowances to cover shortfalls, leading to market instability.

K-ETS applies strict guidelines for new entrants and the removal process. EITE sectors receive 100% of their allowances for free through Phase 3 subject to a condition.³⁰ Here, both banking and borrowing are allowed. The ability to bank and borrow allowances provides

²⁹ The Act included powers for the minister of finance to manage New Zealand's holdings of units that represent New Zealand's target allocation for GHG emissions under the Protocol. In 2019, the Act was amended by the Climate Change Response (Zero Carbon) Amendment Act. These amendments provide a framework by which New Zealand can develop and implement clear and stable climate change policies.

³⁰ $\text{Production cost} \times \text{Trade Intensity} \geq 0.2\%$.

consistent and stable rules that participants can rely on. This regulatory stability reduces the risk of sudden policy shifts that could disrupt long-term planning and investment. Furthermore, by allowing entities to save unused emission allowances (Korean Allowance Units, or KAUs) for future use, banking enables long-term planning. Companies can strategically manage their emissions reductions over multiple compliance periods, reducing uncertainty about future obligations.

The PRC's ETS has yet to establish clear banking regulations. Borrowing was prohibited at first but, later, businesses with a 10% or greater shortfall could seek to borrow up to 50% of the shortfall from a pre-approved allocation for 2023. The inability to bank allowances limits companies' flexibility in managing their compliance strategies, and could lead to inefficient and more expensive compliance pathways.

Robustness

NZ ETS operates with three price management measures: an auction reserve price,³¹ a confidential reserve price,³² and a cost containment reserve (CCR) price.³³ After delinking from the Kyoto market in mid-2015, participants shifted to surrendering NZUs. As domestic emissions prices rose toward the level of the fixed price option, more participants opted to use that option (Leining 2022). The volume of allowances in the CCR can be adjusted based on market conditions, providing flexibility to manage price volatility effectively. The CCR releases additional NZUs if the price exceeds a certain threshold, and therefore can prevent excessive price spikes and provide a predictable upper limit on carbon prices. Furthermore, the flexibility in reserve prices for allowance allocation by auction creates uncertainty.

KazETS does not provide a mechanism such as a price ceiling or floor to adjust price or supply. Meanwhile, the national offset market is poorly developed (Howie, et al. 2020). A weak offset market implies there are fewer incentives to promote investments in innovative and additional emissions reduction projects, such as renewable energy, energy efficiency, and reforestation. This could slow overall progress in emissions reductions.

³¹ An auction reserve price (a form of price floor) sets a minimum price at which the government will sell units at auction.

³² A confidential reserve price at auction is set by the finance ministry based on prevailing prices in the secondary market and helps align auction prices with those in the secondary market.

³³ A CCR (a form of price ceiling) holds a fixed volume of NZUs that are released if the CCR trigger price is reached in the auction.

K-ETS measures for market stabilization,³⁴ market creation, and liquidity management are set outside the total amount of emission allowances, to stabilize and activate the emissions trading market. Mechanisms like offset credits provide flexibility for compliance, although stability in regulatory frameworks and clearer long-term policy signals could strengthen robustness. Companies are allowed to use offsets for up to 10% of their compliance obligations and have developed a national offset market. The K-ETS targets are rigid and do not allow for any change in the national abatement target. In addition, two separate plans were put in place: the 10-year master plan and the commitment period allocation plan. The master plan provides medium- to long-term objectives such as emission targets, whereas the allocation plan defines such elements as the cap; the number of allowances allocated to covered sectors; and rules for banking, borrowing and offsets.

The PRC's ETS operates a market-regulating and protective mechanism created in May 2021 by the Ministry of Ecology and Environment (MEE). This makes it possible for MEE to react to unusual swings in trading prices by, for example, holding an auction, buying back shares, or changing regulations governing the use of China Certified Emission Reductions, but it still needs to describe the mechanism's characteristics and required triggers.

Accountability and Transparency

NZ ETS maintains accountability through clear rules and regulations outlined in the Climate Change Response Act 2002, which establishes compliance obligations and penalties for noncompliance. NZ ETS takes a "self-assessment" approach to MRV of emissions, as in the income tax system. Transparency in NZ ETS ensures that information related to emissions trading, allowance allocations, market transactions, and policy decisions is openly available to the public and stakeholders, which fosters trust (MfE n.d.). Penalties for noncompliance can be imposed, ensuring that participants follow the regulatory requirements and report emissions accurately. In addition to displaying regulatory control, transparent penalty enforcement promotes adherence to emission targets and system integrity. Setting up an effective MRV system has been a challenge. The integrity of the scheme depends critically on accurate self-assessment, regular audits, and the enforcement of penalties for noncompliance. The administrative burden and the need for strong enforcement mechanisms can pose a challenge.

³⁴ Measures include additional auctioning of up to 25% allowances from the market stabilization reserve, the establishment of a limit to the number of allowances entities can hold, an increase or decrease of the borrowing and offset limit, and temporary setup of a price ceiling or price floor (ICAP factsheet on K-ETS).

KazETS suffers from low accountability and transparency, characterized by inconsistent reporting, inadequate monitoring, weak enforcement of penalties for noncompliance (the noncompliance penalty is about \$40/tCO₂), and limited public and stakeholder engagement. These issues undermine trust in the system and hinder its effectiveness in reducing emissions.

K-ETS operates a procedure whereby emission reports are reviewed and certified by the Certification Committee of the Ministry of Environment by the end of May each year. Liable entities are required to revise and resubmit emission reports that are found to be incorrect. Monitoring applies international standards and there is a penalty for noncompliance, which is three times the average market price per tCO₂ for that year to a maximum of about \$91 per tCO₂. Therefore, K-ETS excels in accountability and transparency, with rigorous MRV systems. Public access to emissions data (i.e., NGMS, ETRS, ORS, and KRX), allowance allocations, and market transactions enhances trust and compliance, ensuring the scheme's integrity. The reported data gains credibility through independent verification by qualified experts, lowering the possibility of inaccuracy or manipulation.

The PRC's ETS is at an early stage and so far lacks strong accountability and transparency. As this is a new program, there may be initial challenges in establishing robust MRV systems that ensure accurate emissions data and credible market transactions. The absence of clear reporting protocols and standardized procedures for data disclosure may hinder external verification efforts, leading to uncertainties about the reliability of reported emissions reductions. Moreover, regulatory frameworks and enforcement mechanisms may still be evolving, affecting the consistency and effectiveness of compliance measures. Limited stakeholder engagement and public transparency in the development and implementation phases may also contribute to perceptions of opacity within the ETS. Overcoming these early hurdles will require concerted efforts to strengthen MRV systems, enhance regulatory oversight, and foster greater transparency in operations and decision-making processes to build trust and credibility in the PRC's evolving ETS.

Administrative Cost-Effectiveness

NZ ETS demonstrates administrative cost-efficiency with its reliance on market mechanisms like auctions and trading that minimize direct administrative costs. Robust reporting frameworks ensure compliance without excessive bureaucracy. The ETS employs a combination of upstream and downstream regulatory approaches to effectively manage emissions across various sectors. On the one hand, upstream regulations can enhance administrative efficiency and reduce compliance costs when compared with downstream approaches, which involve more dispersed sources. On the other hand, downstream

regulation ensures that emissions reduction efforts are spread across a wider range of activities and sectors, promoting equity in burden-sharing among different industries and stakeholders. More extensive monitoring and enforcement mechanisms may be required to ensure compliance across diverse activities, which may increase cost.

KazETS has a weak MRV system along with weak government administrative capacity, which increases administrative cost (Howie and Atakhanova 2022; Howie and Akhmetov 2024). At the beginning, trades had to be completed through the Caspy Commodity Exchange JSC, which is still the primary trading platform. Phase 3 saw the opening of trading on other exchange platforms that have contracts with JSC Zhasyl Damu, the company that runs the state registration. With Phase 3 making over-the-counter trading of emission allowances possible,³⁵ this has hampered efficient price discovery and liquidity (Howie, et al. 2022).

K-ETS is buoyed by a strong MRV system that reduces costs. Simplified reporting procedures for small-size offset projects (below 100 tCO₂e) are an example of how this works. That noted, some emissions are double-counted as both upstream and downstream facilities are regulated for electricity, which increases cost.

The PRC's ETS faces challenges with administrative cost-effectiveness that owe to its nascent stage and the complexity of implementing a system at massive scale. The lack of streamlined and standardized procedures for MRV increases administrative burdens, as regulators and companies must navigate new and evolving requirements.

Fairness

NZ ETS demonstrates fairness by incorporating mechanisms to mitigate the impact of carbon pricing across different sectors and stakeholders. It includes transitional measures like the free allocation of allowances to trade-exposed industries to prevent carbon leakage and maintain competitiveness. Additionally, the phased approach to the inclusion of sectors ensures that industries adjust gradually to emission reduction obligations, considering their unique challenges. Transparent allocation and auctioning processes promote equity by distributing emission allowances based on historical emissions and future targets. However, ongoing adjustments are necessary to ensure the burden of carbon pricing is proportionate

³⁵ The over-the-counter market is where securities trade through a [broker-dealer network](#) instead of on a centralized exchange like the [New York Stock Exchange](#). Over-the-counter trading can involve stocks, bonds, and [derivatives](#), which are financial contracts that derive their value from an underlying asset such as a [commodity](#).

and equitable across all sectors, promoting a fair transition to a low-carbon economy while achieving climate objectives.

KazETS is yet to be fair as it does not include mechanisms for revenue generation and redistribution. Moreover, consumers do not contribute to CO₂ reduction in proportion to their income.

K-ETS includes measures to increase fairness, such as the free allocation of allowances to energy-intensive industries and consideration of socioeconomic impacts. Strict regulations on energy prices prevent energy providers from passing on costs in higher prices to low-income consumers. Revenues in Phase 3 were generated from 3% of the allowances auctioned. Revenues from auctioning go into the Climate Response Fund, which supports emissions mitigation infrastructure, low-carbon innovation, and technology development for small- and mid-sized companies covered by the K-ETS.

Under the PRC's ETS, there is currently no arrangement for the use of revenues generated by the scheme.

Tailoring to National Circumstances and Compatibility with Other Policies

NZ ETS is designed to complement national climate policies, such as the Zero Carbon Act 2019, which sets long-term emissions reduction targets. It aligns with New Zealand's commitments under the Paris Agreement and supports potential future linkages with international carbon markets. It is integrated with other domestic environmental and economic policies, ensuring a coherent approach to climate action. Additionally, since agriculture is vital to the country's economy, it is critical to develop new laws or modify current ones to account for agricultural emissions.

KazETS is not sufficiently integrated with existing national policies and strategies for economic development and environmental protection. This leads to potential policy conflicts and inefficiencies.

K-ETS is well tailored to the industrial structure and emission profile, with sector-specific regulations and allowances. The Korean government has shared experiences on carbon pricing with EU-ETS authorities. This has helped ensure policies properly represent national realities while working for emissions reductions in line with economic development goals. Furthermore, it aligns with the ROK's broader climate and energy policies, including its 2050 carbon neutrality target and renewable energy goals.

The PRC's ETS is less tailored to national circumstances and less compatible with other GHG mitigation policies given that its initial focus on the power sector has not fully considered the diverse industrial landscape and regional disparities. The scheme's design may not yet fully integrate with existing national policies and local regulations aimed at GHG reduction, leading to potential overlaps or gaps in policy enforcement.

Summary of Comparisons

Discussion of the eight criteria across the four ETSs is further summarized in Table 4a and Table 4b. One can conclude that, while NZ ETS scores high for most of the criteria, K-ETS comes a close second, followed by the PRC and Kazakhstan, where the former does better in terms of predictability (particularly for indicators on changes in ETS rules and allocating allowances) and robustness (particularly for indicators on use of offsets for compliance). However, it should be noted that all ETSs in this paper remain works in progress. They require enhancements to achieve their full potential and to help meet national GHG mitigation commitments in a cost-effective manner.

Table 4a: Detailed Assessment of Emissions Trading Schemes

Criterion	Indicators	NZ ETS	KazETS	K-ETS	PRC ETS
Contribution to mitigation	1. Scope (% of emissions covered, coverage of key emitting sectors, any important sector omitted)	High	Medium	High	Medium
	2. Stringency of emissions cap	High	Low	Medium	Low
	3. Emissions reduction resulting from ETS	High	Low	High	Low
Cost-effectiveness of mitigation	1. Liquidity of ETS market	High	Low	High	High
	2. Permit prices	High	Low	High	Low
	3. EITE sectors exempted	Low	Low	Low	Low
	4. Level of energy price and cost pass-through regulations	High	Low	High	Low
Predictability	1. Abrupt changes in ETS operation, rules, etc.	High	Low	High	Medium
	2. Allocating allowances to new entrants and removing allowances for closures	High	Low	High	High
	3. Banking and borrowing	High	Low	High	Low
	4. Longer commitment periods	High	Low	High	Low
Robustness	1. Use of PSAMs	High	Low	High	Low
	2. Ability to integrate new participants	High	Medium	High	Medium
	3. Use of offsets as a compliance flexibility mechanism	High	Low	High	Medium
Accountability and transparency	1. Compliance with international MRV standards	Medium	Low	High	Low
	2. Severity of penalties for breaches of accountability	High	Low	High	Medium
	3. Independent body oversees allocation and assessment methods	Low	Low	High	Low

Criterion	Indicators	NZ ETS	KazETS	K-ETS	PRC ETS
	4. Availability and third-party monitoring of data	Low	Medium	High	Low
	5. Operating according to well-developed policies	Medium	Low	High	Low
Administrative cost-effectiveness	1. Length of compliance period	High	Low	High	Medium
	2. Transaction costs—e.g., for brokerage services	High	Low	High	Low
	3. Design of ETS (upstream or downstream) and participation thresholds	High	Low	High	Low
	4. Effective MRV and administration	High	Low	High	Low
Fairness	1. Types and shares of different allocation methods	High	Low	High	Low
	2. Level of harmonization of allocation methodologies	High	Low	High	Low
	3. Period for averaging historical emissions if grandparenting is used	High	N/a	High	N/a
	4. Programs for low-income consumers	N/A	Medium	High	Medium
	5. Vertical equity characteristics of contribution to GHG reduction	High	Medium	High	Low
Tailoring to national circumstances and compatibility with other policies for GHG mitigation and more generally	1. Role of ETS in the climate policy mix	High	Low	High	Medium
	2. Impact of companion policies on ETS outcomes	High	Low	Medium	Low
	3. Impact of ETS on attainment of companion policy objectives	High	Low	Medium	Low

EITE = energy-intensive and trade-exposed, ETS = emissions trading system, GHG = greenhouse gas, MRV = monitoring, reporting, and verification, n/a = not applicable, PSAM = price or supply adjustment measures.

Source: Compiled by authors based on qualitative assessment.

Table 4b: Overall Assessment of Emissions Trading Schemes

Success criterion	NZ ETS	KazETS	K-ETS	PRC ETS
Contribution to mitigation	High	Low	High	Low
Cost-effectiveness of mitigation	High	Low	Medium	Low
Predictability	High	Low	High	Medium
Robustness	High	Medium	High	Medium
Accountability and transparency	Medium	Low	High	Low
Administrative cost-effectiveness	High	Low	High	Low
Fairness	High	Medium	High	Medium
Tailoring to national circumstances and compatibility with other policies for GHG mitigation and more generally	High	Low	Medium	Low

ETS = emissions trading system, GHG = greenhouse gas.

Source: Compiled by authors based on qualitative assessment.

V. CONCLUSION AND WAY FORWARD

This paper has compared six national ETSs now in operation in Asia, in Australia, the PRC, Indonesia, Kazakhstan, the ROK, and New Zealand. Though each is different in ambition, scope, regulations, and efficacy, these ETSs provide useful lessons. The sharing of experiences shows what has worked and what could be done better or differently. This is useful not only for countries with a national ETS in place (or currently being implemented) but also where emissions trading is being considered. India, Malaysia, Pakistan, Taipei, China, Thailand, and Viet Nam are all contemplating setting up versions of an ETS. The paper has also provided a qualitative assessment of four national ETSs: for the PRC, Kazakhstan, the ROK, and New Zealand. These had sufficient information available to prompt useful discussion. It should be noted that each ETS is continuously evolving, and the discussion in the paper refers to conditions until mid-2024.

An ETS is a key policy tool for pricing GHGs and to internalize the global negative externality of emissions. In contrast with taxation on emissions, an ETS helps provide a measurable upper limit on emissions (cap-and-trade), which in turn helps predetermine GHG mitigation. It should be noted that by itself an ETS does not reduce emissions; it simply places an upper bound on them. In response, firms can switch to renewables, adopt cleaner technologies, or use more efficient methods of production.

ETSs have been used successfully to reduce several pollutants across the world. The EU has been using an ETS for GHG mitigation with good effect for nearly two decades—the cap has kept declining and the EU is on track to reduce its emissions by 62% by 2030 compared with 2005 (European Commission 2023). In addition to the EU ETS, there are successful examples in North America of subnational ETSs, such as the Regional Greenhouse Gas Initiative, which has achieved significant reductions in CO₂ emissions from the power sector and has generated revenue for investments in clean energy as well as energy efficiency programs (Natural Resources Council of Maine n.d.). Pricing mechanisms are becoming more commonly used to reduce GHG emissions while interest in ETS is increasing globally.

Almost all the Asian ETSs discussed in this paper are works in progress in design and implementation. Indeed, all countries have a timebound goal of net zero (whether 2050 or 2060) and regard their ETS as central to their GHG mitigation targets. In several instances, however, the coverage of sectors/sources is not yet comprehensive. Exemptions are made for specific industries/sectors, while the caps have been loose and allocations to sources have been generous. Consequently, the price signal has been weak. Managing price volatility and

liquidity in markets for allowances has been an issue. And in their current forms, the monitoring, verification, and enforcement of emissions pose challenges, even as they are evolving.

There are other challenges to tackle. First, the power/energy sector—a major source of emissions that often comes under the ETS—happens to be regulated in nearly all countries. Thus, there is inadequate pass-through of costs, which limits the effectiveness of ETSs.³⁶ Second, the existence of multiple other policies for mitigation also limits effectiveness. Policies include renewable energy mandates, feed-in tariffs, the promotion of energy efficiency, subsidies for fossil fuels, and more. There can also exist overlapping regulations, with layering of ETS over command-and-control regulations.

Asian ETSs operate in the real world and, as such, must take an accommodative approach in the national context. In their current form, these systems are far removed from a textbook “first best” arrangement. On the one hand, one can argue that these ETSs are works in progress and will gradually move closer toward “first best,” where they maximize cost-effectiveness and generate high enough prices to incentivize investment in clean technologies and so forth. The benefits of learning by doing can also be considered, as this can help improve the efficacy of an ETS. On the other hand, one could also argue that the policy objective will continue to be *satisficing* (to produce an adequate result rather than push for an optimal solution). Thus, the ETSs may continue to be improved and their associated policies revised while continuing to exist in an ecosystem with multiple other carbon policies. This implies that ETSs will contribute to mitigation or carbon neutrality promises but only alongside other policies that countries feel are necessary and just. At this juncture, it is difficult to provide a definitive answer on this.

This paper puts forward the following recommendations to improve ETS predictability and robustness with a view to helping firms make long-term investment decisions.

- The objectives in implementing an ETS should be clearly laid out to make it clear whether the cap will be the main instrument for GHG mitigation and achieving net zero. If not, the role of ETS in the overall strategy for attaining carbon neutrality should be determined.
- Long-term plans for an ETS should be clear. These may include decisions and timelines on bringing in additional sectors and phasing out exemptions for industries

³⁶ See, for example, Acworth, et al. (2019) and Yu, et al. (2023) for an opposing view.

such as EITE. It should also be decided whether the cap needs to be progressively tightened.

- PSAMs should be considered as a tool to manage liquidity and prices in the market.
- While most ETSs use emission intensity benchmarks and therefore the cap is bottom-up or endogenous, there should be a timeframe to move to mass-based standards—i.e., a fixed top-down cap that is grandparented and/or auctioned.
- Administrative capacity, especially for MRV, is critical to the success of ETS and should be strengthened over time. Steps may include use of third-party verification, application of technology for remote monitoring, and so forth.
- Stakeholders should be consulted in both the design and the implementation stages of an ETS, to create “buy in” and enhance acceptability. Specifically, attaining the legitimacy of an ETS from a business perspective is crucial.
- Developing and emerging market economies need to adopt complementary policies along with their ETS to address the challenges related their energy sector. Policies such as targeted subsidies can help mitigate the impact of higher energy prices arising as a result of carbon pricing, and thus manage the associated political and economic fallout. By implementing supportive measures, governments can effectively balance carbon reduction goals and distributional concerns, and avoid significant economic disruptions and political resistance.

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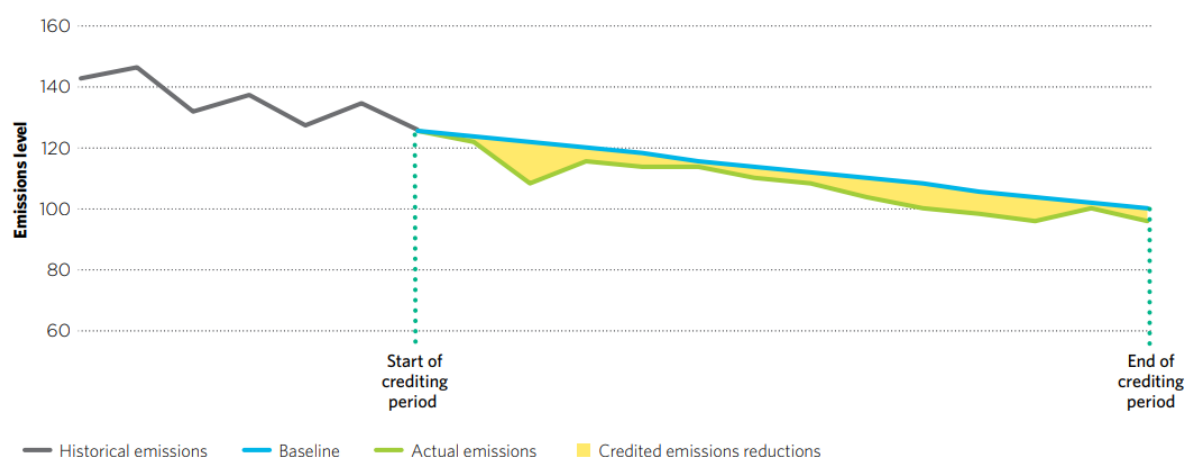
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APPENDIX 1: ABOUT THE NATIONAL EMISSIONS TRADING SCHEMES

Australia does not have a formal ETS. Instead, it has had a baseline-and-credit system since 2023. Unlike an ETS, this has no emission caps. Firms below a given baseline can generate and sell credits to other firms that exceed the baseline (Appendix Figure 1). A scheme known as the Safeguard Mechanism, introduced in 2016, assigns mandatory intensity-based emissions baselines for over 200 large industrial emitters in Australia. This scheme became a baseline-and-credit system only in July 2023, after the Labor government³⁷ issued Safeguard Mechanism Credits (SMCs) to facilities that overachieved against their baseline. This in effect turned the Safeguard Mechanism into a mandatory baseline-and-credit system, with a fixed yearly decline rate for facility-level baselines. Those emitting above their baseline must offset excess emissions by surrendering SMCs or Australian Carbon Credit Units (ACCUs).³⁸ The introduction of SMCs was accompanied by a tightening of baselines and a default decline rate of 4.9% per year to align the outcome with Australia's 2030 targets.

Figure A1: Offsets Baseline and Credit Scheme



Source: Climate Change Authority. 2014. [Coverage, Additionality and Baselines—Lessons from the Carbon Farming Initiative and Other Schemes CCA Study](#).

The PRC launched its national **ETS** in 2021, which (as of now) covers only gas- and coal-fired power plants.³⁹ These produce about two-thirds of electricity in the PRC. The ETS covers approximately 2,200 firms whose annual emissions exceed 26,000 tCO₂. Most of the regulated

³⁷ Climate policy in Australia is highly politicized, especially where related to carbon pricing, with policies supported by the Labor Party and opposed by the Liberal Party. Several flip-flops on climate change, and carbon pricing in particular, have occurred as governments change. See Crowley (2017).

³⁸ Both are both tradable financial products representing 1 tCO_{2e} emissions. Both units are managed by the Clean Energy Regulator and held in a business' Australian National Registry of Emissions Units account. ACCUs have been around since 2015 under a different scheme. For details, see Carbon Market Institute (2023).

³⁹ The scope of the PRC's ETS is expected to be gradually expanded to cover seven other sectors (petrochemicals, chemicals, building materials, steel, nonferrous metals, paper, and domestic aviation). Entities in these sectors have had MRV obligations since 2015. There is no specific timeline for this expansion.

entities are state-owned enterprises. This is the world's biggest ETS in terms of the volume of GHGs regulated but not yet by value or volume of trading. The national ETS was preceded by pilot ETSs in several cities/regions. The pilots continue to function concurrently with the national ETS, encompassing industries and entities that are not covered by the national system. As the latter's coverage increases, entities currently covered by the pilots will be incorporated into the national ETS. The most notable feature of the PRC's ETS is that the cap is not fixed but is determined bottom-up by aggregating the allowances distributed to all firms. This in turn depends on a firm's current and last year's output times a benchmark emissions intensity. In effect, the ETS is a tradable performance standard, a rate-based instrument that is different from cap-and-trade. A firm can influence its allowance allocation through its choice of output during the compliance period.

MRV remains an important concern for the PRC's ETS. Officials from the provincial level of MEE are sent to carry out inspections but some provinces have inadequate personnel and technical ability (Sandalow, et al. 2022). They also face a conflict of interests, as many of the biggest emitters are state-owned companies that are critical to local growth, development, and the security of energy supplies (ibid.). Furthermore in, March 2022, the national MEE discovered problems with emissions data submitted by power plants following an inspection campaign (Bloomberg 2022; Reuters 2022). As a result regulations were tightened in February 2024 (to take effect from May 2024) and significantly increased the punishment for noncompliance, data fraud, and market manipulation. The change also elevates the governance of the ETS to the State Council, up from its previous ministerial oversight (ICAP 2024).

Indonesia is an economy rich in fossil fuels and minerals. Over the past 20 years, the economy has been characterized by dynamic growth, with massive increases in electricity demand, coal consumption, and exports. The country is the world's fifth largest producer of coal as well as the biggest exporter (\$104 billion in coal exports in 2023). Hence, GHG emissions are rising steadily, pushing Indonesia's global ranking into sixth place, just ahead of Japan. Coal continues to be a major challenge for Indonesia's GHG mitigation efforts. A huge fleet of newly operating off-grid coal plants are fueling the booming domestic industry in metals (such as nickel and copper)⁴⁰ and large infrastructure projects for the new capital city.⁴¹

⁴⁰ Indonesia has become the world's largest nickel producer within just a few years, as its share in global nickel extraction grew from 5% in 2015 to 50% in 2023. The country is also the world's third copper ore exporter.

⁴¹ "In November 2023, the JETP [Just Energy Transition Partnership] Comprehensive Investment and Policy Plan (CIPP) communicated that new insights on the huge pipeline of off-grid coal plants would render Indonesia's initial target to cap power emissions at 290 MtCO₂ in 2030 'infeasible'. The plan now aims to cap on-grid emissions at 250 MtCO₂ in 2030. Off-grid coal could reach 30 GW in 2030 and result in an additional 150 MtCO₂,

The Indonesia ETS (known as **NEK**)⁴² is a mandatory, intensity-based ETS for the power sector, launched in early 2023. The first phase (2023–2024) will be followed by two more (2025–2027 and 2028–2030). All these phases apply exclusively to the power sector. Initially, NEK focuses on 99 coal-fired grid-connected power plants with a capacity of 25 MW or more. Most of the plants covered are owned by the state-owned electricity company Perusahaan Listrik Negara. The government establishes intensity targets (known as PTBAE) that determine the number of allowances installations receive for each MWh of electricity generated. Covered entities are required to surrender allowances corresponding to all their covered emissions. Additionally, they have the option to purchase allowances through auctions. The Indonesian Carbon Exchange (IDXCarbon) started in September 2023. Given the recent nature of this start, no data is available on verified emissions for the first period of the ETS. Eventually the ETS will function as a hybrid “cap-tax-and-trade” system, operating concurrently with a carbon tax to be introduced around 2025. Facilities failing to meet their obligations under the ETS will be subject to this tax, whose rate will be aligned with the domestic carbon market’s price.

Kazakhstan launched its ETS (**KazETS**) in January 2013. This covers half of the country’s CO₂ emissions from power, centralized heating, extracting industries, and manufacturing sectors. Unlike in NZ ETS, the KazETS cap is formed bottom-up, based on installations’ expected production output and a benchmark. A reserve of allowances, on top of the cap, is further available to covered entities in case they exceed their planned output. KazETS entered its 12th year of operation in 2024 and work is underway to introduce auctions, while benchmarking is the only method of allocating allowances since 2021.

Major factors that have had impacts on the performance of KazETS are as follows:⁴³ (i) energy prices in Kazakhstan are regulated and producers are not able to pass on increased compliance costs to end users;⁴⁴ (ii) most of the installed capacity in both electricity and district heat is controlled by state-owned enterprises that are not cost minimizers; (iii) the allowance market is oligopolistic with a few dominant firms and a competitive fringe;⁴⁵ (iv) during Phase 2 (2014–2015), some firms received additional allowances if they introduced a new source of emissions or increased capacity;⁴⁶ while (v) during Phase 2, firms had to conduct trading

meaning total power sector emissions could reach 400 MtCO₂ in 2030, more than twice the level required to align with the Paris Agreement’s 1.5C temperature limit” ([Climate Action Tracker](#), accessed 6 April 2024).

⁴² Nilai Ekonomi Karbon, or Economic Value of Carbon Trading Scheme in English.

⁴³ This discussion is based on Howie, et al. (2020).

⁴⁴ All changes to electricity and heat tariffs must be submitted to the Kazakhstan Antimonopoly Commission for approval. The cost of carbon is not on the list of cost elements and therefore cannot be passed on to consumers.

⁴⁵ In Phase 3, three large industrial groups (Eurasian Resources Group, Samruk Energy, and Arcelor Mittal) together received 49% of allowances.

⁴⁶ 4 million additional allowances were issued outside of the allowance reserve, in effect increasing the cap.

through a designated carbon exchange (Caspy), where large brokers acted both as buyers and as sellers, which resulted in non-transparent pricing; and finally (vi) major currency devaluations increased the financial burden on companies with foreign currency loans and made industry associations lobby against KazETS.

The **ROK's** Emissions Trading Scheme (**K-ETS**), launched in 2015, is East Asia's first nationwide, mandatory ETS.⁴⁷ It covers around 89% of the ROK's national GHG emissions and will help the country in its objective to become carbon-neutral by 2050 (ADB 2018). Allocation is mostly free (through very limited auctions) and based on historical emissions or output. EITE industries get free allocation.

A notable feature of K-ETS is that it covers both direct and indirect emissions (but only those from electricity consumption).⁴⁸ The latter is achieved by adding the cost of carbon to the final consumer price and is used to circumvent incomplete pass-through of carbon prices owing to government regulation. Another interesting feature is the role of officially designated market makers. These are third-party participants who can inject liquidity (i.e., allowances) into the market (Yu nd). Finally, K-ETS gives special consideration to EITE industries (as in NZ ETS).

The **New Zealand** ETS (**NZ ETS**) was launched in 2008 and is the main policy for GHG mitigation. In 2021, it covered 37 MtCO₂e (48%) of GHG emissions. The cap is top-down and aligned with the country's 2050 net zero target and associated emissions budget. Covered entities must surrender allowances for all their reported emissions. The ETS has a broad coverage of sectors, including forestry, energy, industrial processing, waste, and more (Tao, et al. 2024). Allocation is based primarily on auctioning. Free allocation is granted for EITE activities. Three notable features of NZETS are (i) inclusion of the forestry sector, which can earn allowances for emission removals (also known as forestry removals),⁴⁹ (ii) exclusion of emissions from agriculture, despite the sector's 51% share of GHG emissions, the highest among all Organisation for Economic Co-operation and Development countries (Liao, et al. 2023),⁵⁰ and (iii) the fact that, to date, it is the only ETS that operated with buy-and-sell linkages to the Kyoto market, from 2008 through mid-2015 when it was delinked (for details see Leining and Kerr 2016; Leining, Kerr, and Bruce-Brand 2019).

⁴⁷ The discussion of K-ETS draws on Yu (2023) and Yu (nd).

⁴⁸ For a critique of this approach, see Kim and Lim (2020).

⁴⁹ Participants are entitled to receive one allowance (aka NZU) per ton of CO₂ removed for registered post-1989 forest land. A total of 12 million NZUs were issued for forest removal activities for the 2022–2023 reporting year ([ICAP factsheet on New Zealand ETS](#), accessed on 25 June 2024).

⁵⁰ Though NZ ETS was envisaged as covering "all sectors and all gases," it didn't quite work out that way. Biogenic methane emissions (from sheep and cattle) that are counted under agricultural GHG emissions have been kept out of the ETS. In fact, they are not regulated at all at present.

APPENDIX 2: DESCRIPTION OF EACH OF THE CRITERIA

Contribution to mitigation

This refers to the degree to which the ETS has achieved the outcomes compared with the intended goals of reducing emissions.

Cost-effectiveness of mitigation

This is defined as achieving the emissions target by the cheapest means possible. The greater the flexibility on when and where emission reductions take place, the higher the potential for low-cost emissions reductions. Greater cost-effectiveness enhances the contribution of an ETS to mitigation. At the same time, the ability of an ETS to deliver least-cost abatement across covered sectors is influenced by how well it is integrated with other policies that affect emissions (e.g., energy sector policies).

Predictability

Predictability is improved when key features of the ETS are communicated effectively and clear processes and parameters for changes are provided, as well as by avoiding sudden change.

Robustness

This refers to the effectiveness of mechanisms to manage price and quantity shocks. While some variability in prices is to be expected, and is desirable as a signal to market participants, excessive price volatility resulting from exogenous shocks, regulatory uncertainty, and market imperfections necessitates intervention in the market. Correct judgment about the acceptable level of variability and effective functioning of PSAMs⁵¹ ensure a consistent price signal for investment and a more robust ETS. Robustness, however, should not be mistaken for ad hoc intervention, nor is it in contradiction with the predictability criterion.

Accountability and transparency

Strong MRV and enforcement and robust registry design ensure accountability and transparency of the ETS. Design decisions must also be made transparently to help build trust in the system to allow market participants and investors to plan their business.

Administrative cost-effectiveness

⁵¹ Examples of PSAMs to address low prices include auction reserve prices, hard price floors, and levying additional fees and charges on top of the allowance price. PSAMs to address high prices include cost containment reserves and hard price ceilings.

Administrative costs depend on the scope of the system, the choice of point of obligation, the frequency with which data needs to be reported and compliance proven, and the requirements for compliance and enforcement. The tradeoff is between reducing transaction costs and achieving ideal accountability and transparency outcomes, particularly with respect to MRV.

Fairness

Emissions trading is not possible without mechanisms accepted by stakeholders and political support. Ensuring fairness for all involved, especially in the distribution of costs and benefits, is essential for gaining and maintaining support and ensuring the ETS will endure.

Tailoring to national circumstances and compatibility with other mitigation policies

It is important to align ETS mechanisms with the existing regulatory environment, trade and competitiveness concerns, and institutional strengths and weaknesses.⁵² It is also important to examine compatibility with other mitigation policies (including where it is lacking) and to manage policy interactions: ETSS operate within broader policy and legal frameworks, including other GHG mitigation policies (both market and non-market) such as carbon taxes, subsidies for renewables and energy efficiency, and many more. These policies can either counteract/distort/duplicate the impact of an ETS or complement it and make it more effective (e.g., by creating additional GHG mitigation opportunities or by removing non-price barriers to reduce emissions).

⁵² See, for instance, ADB (2016).